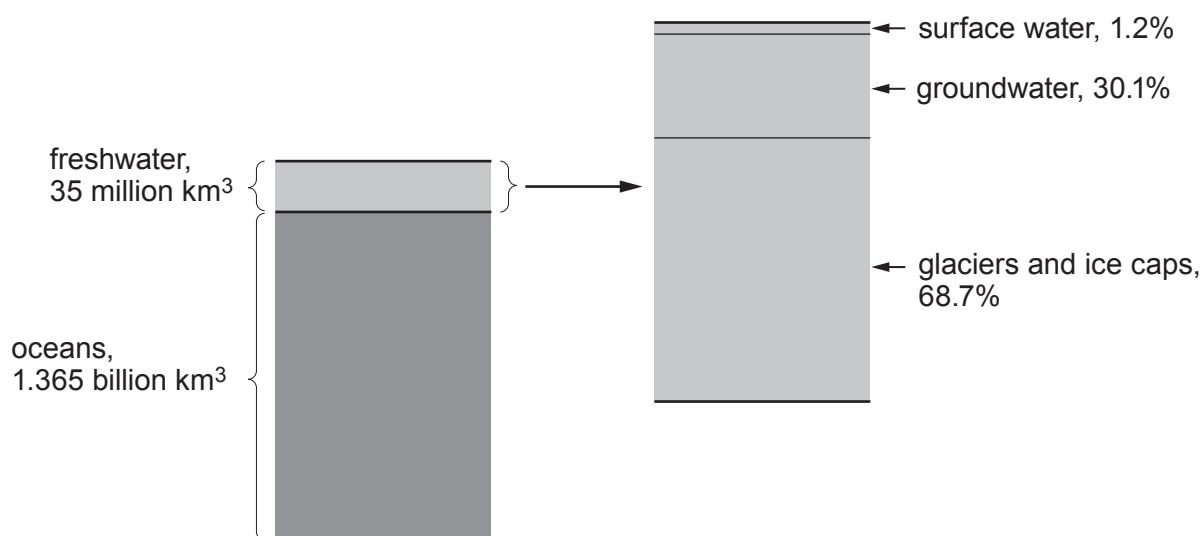


8. The diagram shows the distribution of water on Earth.



- (a) Use the information in the diagram to calculate the volume of freshwater that is available as surface water. [2]

Volume = million km³

- (b) As the Earth's population continues to increase, there is going to be a greater need to get freshwater from seawater. This is known as desalination.

Give a method that could be used to obtain freshwater from seawater. [1]

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- (c) Calcium ions, Ca^{2+} , cause hardness in groundwater.

Describe an experiment to **compare** the hardness of two groundwater samples. [3]

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- (d) Washing soda is used to soften hard water. This contains sodium carbonate.

Give the formula of sodium carbonate. [1]

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END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		





THE PERIODIC TABLE

Group

1 2

3

4

5

6

7

0

1

H

Hydrogen

1

1H Hydrogen 1																				4He Helium 2																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																															

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number